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Heat deflector

Abstract

A heat deflector is disclosed that includes a first portion having a first outer edge and a center and a first surface extending from the first outer edge to the center of the first portion. The first surface sloping downward from the first outer edge toward the center of the first portion. The heat deflector further includes a second portion having an inner edge and an outer edge. The inner edge of the second portion is adjacent to the first outer edge of the first portion. The heat deflector further includes a third portion having a third outer edge and a third inner edge and a second surface extending from the third outer edge of the third portion to the third inner edge of the third portion. The second surface sloping downward from the third outer edge of the third portion to the third inner edge of the third portion.

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Background/Summary

TECHNICAL FIELD

BACKGROUND

(3) Large fire pits are generally fueled by piles of logs or combustible materials. Heat from the fire pit rises mostly upward with some lateral

radiation of heat. A heat deflector may therefore be placed over the fire pit to increase the lateral radiation of heat from the fire pit so that people sitting around the firepit can receive some of the heat. Current heat deflector designs may be large and heavy, making them challenging to move.

Furthermore, current designs may not effectively redirect the heat of the fire pit. Additionally, current designs may reduce the efficiency of the fire pit.

(4) It is therefore to be appreciated that such commonly used heat deflectors could be improved. Accordingly, a need exists for heat deflectors that improve the performance and efficiency of deflecting the heat generated by the fire pit that address the forgoing and other concerns.

SUMMARY

(6) One general aspect includes heat deflector including a first portion having a first outer edge and a center, and having a first surface extending

from the first outer edge to the center of the first portion. The first surface may slope downward from the first outer edge toward the center of the first portion. A second portion may have a second inner edge and a second outer edge. The second inner edge of the second portion may be adjacent to the first outer edge of the first portion. A third portion may have a third outer edge and a third inner edge. The heat deflector may have a second surface extending from the third outer edge of the third portion to the third inner edge of the third portion. The second surface may slope downward from the third outer edge of the third portion to the third inner edge of the third portion.

(7) Some example implementations may include one or more of the following features. In some embodiments, the heat deflector further includes a plurality of holes formed in the second portion, the plurality of holes sized to permit passage of air through the second portion. In some embodiments, the heat deflector further includes one or more ribbed features formed in the third portion, the one or more ribbed being configured to improve resistance of the heat deflector to deformation. In some embodiments, the heat deflector further includes one or more legs, a top portion of each of the one or more legs being shaped to pass through the second portion and secure to the heat deflector, each of the legs being removable from

the heat deflector. In some embodiments, the heat deflector further includes a hole formed in the center of the first portion. In some embodiments, the slope of the first portion has a first angle, and the slope of the third portion has a second angle that is smaller than the first angle. In some embodiments, the first portion has a generally circular shape.

(8) Another general aspect may include a heat deflector including a cone having a base perimeter, a vertex, and a first surface extending from the vertex to the base perimeter, wherein the vertex is below the base perimeter. The heat deflector also may include a first ring extending around the base perimeter, with the first ring having an inner perimeter and an outer perimeter. A plurality of holes may be formed in the first ring. A second ring may extend around the first ring. The second ring may have a second surface that has a first slope extending from an interior edge of the second ring to an exterior edge of the second ring.

(9) Some example implementations may include one or more of the following features. In some embodiments, the first surface has a second slope that is greater than the first slope of the second ring. In some embodiments, the heat deflector further includes a third ring extending from the first ring to the second ring, the third ring including a raised surface; an opening in the third ring; and a leg having an upper portion, the upper portion of the leg extending through the opening in the third ring. In some embodiments, the heat deflector further includes one or more structural support features formed in the second surface of the second ring. In some embodiments, the heat deflector further includes a hole formed in vertex of the cone. In some embodiments, the heat deflector is formed of a single stamped piece of metal. In some embodiments, the cone and the second ring are operable to laterally deflect upward radiating heat. In some embodiments, the heat deflector further includes a third ring extending from the first ring to the second ring, the third ring including a raised surface, wherein the raised surface of the third ring receives a support post for installing the heat deflector.

(10) Yet another general aspect includes a heat deflector including a plurality of concentric rings and a cone located within an inner ring of the plurality of concentric rings. The cone may have a first surface with a first slope from the inner ring to a vertex of the conc. An outermost ring of the plurality of concentric rings may have a second surface. The second surface may have a second slope from an outer edge of the outermost ring to toward the center of the outermost ring. The first slope may be greater than the second slope. A leg may removably attach to one of the inner ring and the outermost ring.

(11) Some example implementations may include one or more of the following features. In some embodiments, the heat deflector further includes a plurality of holes formed in a middle ring of the plurality of rings, the middle ring being between the inner ring and the outermost ring. In some embodiments, the heat deflector further includes a cap to secure the leg to the heat deflector. In some embodiments, the heat deflector further includes a raised ring of the plurality of concentric rings, the raised ring being between the inner ring and the outermost ring; and an opening in a sidewall of the raised ring through which the leg is attachable to the heat deflector. In some embodiments, the heat deflector further includes one or more structural features formed in the outermost ring to resist warping the heat deflector during heating and cooling cycles.

(12) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to limit the scope of the claimed subject matter. A more extensive presentation of features, details, utilities, and advantages of the heat deflector, as defined in the claims, is provided in the following written description of various embodiments of the disclosure and illustrated in the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Illustrative embodiments of the present disclosure will be described with reference to the accompanying drawings, of which:
- (2) FIGS. 1A and 1B are side and perspective views respectively of an example heat deflector disposed on a combustion fire pit using removable legs, according to some embodiments of the present disclosure.
- (3) FIGS. 2A-2C are side, perspective, and exploded illustrative views respectively of an example heat deflector disposed on a combustion fire pit using a hub, according to some embodiments of the present disclosure.
- (4) FIG. 3 is a side view of an example radiant energy from the heat deflector connected to the combustion fire pit, according to some embodiments of the present disclosure.
- (5) FIGS. 4A-4E are illustrative views of an example heat deflector and removable leg components, according to some embodiments of the present disclosure.
- (6) FIGS. 5A-5C are illustrative side views of attaching a removable leg to a heat deflector, according to some embodiments of the present disclosure.
- (7) FIGS. 6A-6C are illustrative perspective views of securing a removable leg to a heat deflector, according to some embodiments of the present disclosure.
- (8) FIG. 7 is an illustrative bottom view of an example heat deflector, according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

- (9) In accordance with at least one embodiment of the present disclosure, a heat deflector is provided which includes novel structural features to laterally redirect upward moving radiant energy and improve the strength of the heat deflector. The disclosed heat deflector also includes air flow features conducive to reducing back pressure on the fire pit caused by the presence of the heat deflector. The air flow features allow smoke and some heated air to pass upwardly through the heat deflector, which may reduce back pressure and increase the air-flow and therefore increase the heat and energy output from the first pit. Further features provide for portability and modular installation of the heat deflector. In some examples, the heat deflector may be installed on the fire pit through the use of removable legs, a hub, or other components.
- (10) For the purposes of promoting an understanding of the principles of the present disclosure, reference will now be made to the embodiments illustrated in the drawings, and specific language will be used to describe the same. It is nevertheless understood that no limitation to the scope of the disclosure is intended. Any alterations and further modifications to the described devices, systems, and methods, and any further application of the principles of the present disclosure are fully contemplated and included within the present disclosure as would normally occur to one skilled in the art to which the disclosure relates. In particular, it is fully contemplated that the features, components, and/or steps described with respect to one embodiment may be combined with the features, components, and/or steps described with respect to other embodiments of the present disclosure. For the sake of brevity, however, the numerous iterations of these combinations will not be described separately.
- (11) These descriptions are provided for exemplary purposes only, and should not be considered to limit the scope of the heat deflector, removable legs, hub, or fire pit. Certain features may be added, removed, or modified without departing from the spirit of the claimed subject matter.
- (12) FIGS. 1A and 1B are a side view and a perspective view, respectively, of an exemplary heat deflector **102** disposed on an exemplary fire pit using removable legs in accordance with some embodiments of the present disclosure. FIG. 4D shows a cross-sectional view of the main body of the heat deflector **102**. The heat deflector **102** may include removable legs **104**, and is shown on a fire pit **106**. The fire pit **106** may be an example of the fire pit described in U.S. application Ser. No. 17/169,269 entitled “Combustible Fuel Burning Fire Pit with Removable Fire Grate and Ash Pan” filed on Feb. 5, 2021, which is incorporated herein by reference in its entirety. The fire pit **106** includes an interior wall and an exterior wall with an air gap between the interior wall and the exterior wall. With reference to FIGS. 1A and 1B, the fire pit **106** includes a combustion chamber **120** and is divided into a top portion **108**, a middle portion **110**, and a bottom portion **112**. In some example implementations, the fire pit **106** is round or has a cylindrical shape with a diameter or width **D1** that may be about 250 mm to about 800 mm. In some examples, width **D1** may be about 350 mm to about 700 mm. In such implementations, the heat deflector **102** also may be round and may have a diameter or width **D2** that is greater than **D1**.

Depending on the implementation, the width D2 may be about 400 mm to about 900 mm. In some examples, the width D2 may be about 500 mm to about 800 mm. Although larger and smaller sizes of the fire pit **106** and the heat deflector **102** are also contemplated. Also depending on the implementation, the ratio of the width of the heat deflector to the diameter or width of the firepit may be within a range of about 1:1 to 1.3:1. Although larger and smaller ratios are also contemplated.

(13) The top surface **102t** of the heat deflector **102** may be at height HI above a top surface **106t** of the fire pit **106** to maximize the deflection of the radiant energy from the combustion chamber **120** of the fire pit **106**. The height HI may be about 150 mm to about 915 mm. In some examples, the height HI may be about 225 mm to about 750 mm. In some embodiments, the ratio of the of the height HI to the width D2 is in the range of about 2.5:1 to 5:1. Yet other ratios are contemplated.

(14) The top portion **108** of the fire pit **106** includes a connecting ring **114**, a top lip **116**, and a plurality of top ventilation holes (not shown) inside the combustion chamber **120**, along the interior wall of the fire pit **106**. The connecting ring **114** is a horizontal surface extending from the interior wall to the exterior wall of fire pit **106**. The connecting ring **114** may be used to interface with the removable legs of the heat deflector **102** to the fire pit **106**. The top lip **116** is located along the inner surface of connecting ring **114** and adjacent the combustion chamber **120**. The top lip **116** may be a guide to properly place the removable legs **104** of the heat deflector **102**. Additionally, the top lip **116** may protect the heat deflector **102** from falling off the fire pit **106** by preventing the legs **104** from sliding into the combustion chamber **120**. The plurality of top ventilation holes provides air flow for the fire pit and improves the burning of combustible material in the combustion chamber **120** while reducing smoke as described in U.S. application Ser. No. 17/169,269. The plurality of top ventilation holes may be round and punched through the interior sidewall of the fire pit **106**. In some embodiments, the plurality of top ventilation holes may be square, rectangular, oval, or another shape. The plurality of top ventilation holes are connected to a plurality of bottom ventilation holes **115** located in the bottom portion **112** of the fire pit **106**.

(15) The middle portion **110** of the fire pit **106** includes interior and exterior sidewalls of the fire pit **106**. In some implementations, the sidewalls of the fire pit **106** may each be formed of a single sheet of metal. In some embodiments, the sidewalls of the fire pit **106** may be formed of another material. In some embodiments, an air gap exists between the interior sidewall and the exterior sidewall to allow for air to flow from the plurality of bottom ventilation holes **115** to the plurality of top ventilation holes.

(16) The bottom portion **112** of the fire pit **106** includes the plurality of bottom ventilation holes **115**. The plurality of bottom ventilation holes **115** may be round and punched through the exterior sidewall of the fire pit **106**. In some embodiments, the plurality of bottom ventilation holes **115** may be square, rectangular, oval, or another shape.

(17) The heat deflector **102** includes an outer portion **130**, a middle portion **132**, and an inner portion **134**, that together form a main body. In some implementations, the heat deflector **102**'s main body (absent the legs) may be formed from a single stamped sheet of metal, such as for example, a stainless steel sheet. In some embodiments, the main body of the heat deflector **102** may be formed by joining two or more pieces together, such as by the use of welds, rivets, bolts, or other means of joining two metal pieces.

(18) The outer portion **130** extends from an outer edge **131** to an inner edge **133**. A bottom surface of the outer portion **130** slopes downward from the outer edge **131** toward the inner edge **133** at an angle $\alpha 1$, visible in FIG. 1A. In some embodiments, such as when the main body of the heat deflector **102** is formed of a single sheet, the top surface of the outer portion **130** has the same slope of angle $\alpha 1$. The angle $\alpha 1$ may be about 3° to about 12°. In some examples, the angle $\alpha 1$ may be about 5° to about 10°. Although larger and smaller angles are contemplated. In some implementations, the angle $\alpha 1$ of the outer portion **130** may maximize the angle of incidence of the strongest source of radiant energy in relation to a location on the outer portion **130** of the heat deflector **102**, as seen in FIG. 3. For example, fuel burning in the combustion chamber **120** causes radiant energy (e.g., heat) to move upward from the fire pit **106**. The heat deflector **102**, specifically the outer portion **130**, laterally deflects the upward moving radiant energy away from the fire pit **106**. The angle $\alpha 1$ of the outer portion **130** may be selected to maximize or provide high lateral deflection of the radiant energy for each location along the outer portion **130** of the heat deflector **102**.

(19) In the implementation shown, the outer portion **130** of the heat deflector **102** includes one or more ribs or ribbed features **136**. Each rib or ribbed feature **136** may be stamped into the outer portion **130** of the heat deflector. In some embodiments, each ribbed feature **136** may be a separate piece that is joined to the heat deflector through the use of welds, rivets, bolts, or other fastening means. As illustrated, the ribbed feature **136** includes sidewalls **137** and a downward sloping surface having an angle $\alpha 2$ that may be greater than the angle $\alpha 1$ of the outer portion **130** of the heat deflector **102**. The angle $\alpha 2$ may be about 5° to about 15°. In some examples, the angle $\alpha 2$ may be about 7° to about 12°. Although larger and smaller angles are contemplated. In some embodiments, the ribbed features **136** are evenly, or equidistantly, spaced around the heat deflector **102**. In some other embodiments, the ribbed features **136** are unevenly spaced around the heat deflector **102**. In the illustrated embodiments, there are three ribbed features **136** spaced equidistant from each other around the outer portion **130** of the heat deflector **102**. The ribbed features **136** may be structural support features that improve the strength and stability of the heat deflector **102**. The improved strength may increase resistance to warping, deforming, cracking, and/or breaking that may otherwise be caused by the heating and cooling cycles during use of the heat deflector **102**.

(20) The middle portion **132** of the heat deflector **102** extends from the inner edge **133** to an inner edge **135** and includes a raised ring **138** and a plurality of holes **140**. The raised ring **138** is located around the outer periphery of the middle portion **132** along the inner edge **133**. In some embodiments, the raised ring **138** is about halfway between the outer edge **131** and the center of the heat deflector **102**. Here, a top surface **139** of the raised ring **138** is higher than the inner edge **133** of the outer portion **130** of the heat deflector **102**. The raised ring **138** includes connection points **122** for attaching the removable legs **104** to the heat deflector **102**. The connection points **122** may be slots, apertures, openings, or other engagement features that may enable the removable legs **104** to engage and attach to the raised ring **138**. An upper portion **105** of the removable legs **104** attaches to the heat deflector **102** at connection points **122**. A bottom portion **107** of the removable legs **104** attach to the connecting ring **114** of the fire pit **106**. In some embodiments, the removable legs **104** rest on the connecting ring **114** and are held in place by the top lip **116** and the weight of the heat deflector. In other embodiments, the removable legs **104** are attached to the connecting ring **114** through the use of a fastener.

(21) Adjacent to and disposed radially inward from the raised ring **138**, is a surface comprising the plurality of holes **140**. The plurality of holes **140** create an array of passages through the main body of the heat deflector that serves multiple functions. The plurality of holes **140** allow smoke to rise through the heat deflector **102** so that all of the smoke does not have to work its way around the outer edges of the heat deflector **102**. Further, the plurality of holes **140** reduces downward pressure on the convective upward air movement from the combustion chamber **120** that is caused by the heat deflector **102**. The downward pressure, or back pressure effect, reduces the efficiency of the fire pit **106**, and specifically, the combustion chamber **120**. Accordingly, the plurality of holes **140** improves the efficiency of the fire pit **106**. The improved efficiency may be measured in heat produced, fuel consumption rate, smoke produced, etc. Furthermore, the plurality of holes **140** allow liquid (e.g., rain) to drain from the heat deflector **102**. As depicted, the plurality of holes **140** are round in shape. In some embodiments, the plurality of holes **140** may have other shapes. They may be square in shape. In some embodiments, the plurality of holes **140** may be elongated slots. In some embodiments, the plurality of holes **140** may be replaced with a screen that performs the same function. There may be more or fewer holes among the plurality of holes **140** than depicted in FIGS. 1A, 1B, and 4D.

(22) The inner portion **134** of the heat deflector extends from the inner edge **135** to the center of the heat deflector **102**. The inner portion **134** has a cone shape with a base downward extending vertex **141** (FIG. 4D). The vertex **141** of the cone within the inner portion **134** may include one or more holes **142**. The depicted embodiment includes a single hole **142**. The holes **142** allow liquid (e.g., rain) to drain from the inner portion **134** of the heat deflector **102**. The hole **142** may also relieve downward pressure on the convective upward air movement that is caused by the heat deflector **102**. A bottom surface of the inner portion **134** has an angle $\alpha 3$ that is greater than the angle $\alpha 1$ of the outer portion **130**. The angle $\alpha 3$ may be about 20° to about 40°. In some examples, the angle $\alpha 3$ may be about 25° to about 35°. Although larger and smaller angles are contemplated. The angle $\alpha 3$ of the inner portion **134** maximizes the angle of incidence of the strongest source of radiant energy in relation to a location on the inner portion **134** of the

heat deflector **102**, as seen in FIG. 3. For example, fuel burning in the combustion chamber **60** causes radiant energy (e.g., heat) to move upward from the fire pit **106**. The heat deflector **102**, specifically the inner portion **134**, laterally deflects the upward moving radiant energy away from the fire pit **106**. The angle α_3 of the inner portion **134** maximizes the lateral deflection of the radiant energy for each location along the inner portion **134** of the heat deflector **102**.

(23) FIGS. 2A-2C are a side view, a perspective view, and an exploded perspective view, respectively, of an exemplary heat deflector installed on an exemplary fire pit using a central hub in accordance with some embodiments of the present disclosure. FIGS. 2A and 2B illustrate the heat deflector **102** installed on the fire pit **106** using a hub **150** instead of the removable legs **104**, as illustrated in FIGS. 1A and 1B. The fire pit **106** has been described above with respect to FIGS. 1A and 1B. The hub **150** includes a bottom portion **152** and a top portion **154** and supports **156** connecting the bottom portion **152** and the top portion **154**. The bottom portion **152** of the hub **150** has a diameter or width D3 that may be about 250 mm to about 750 mm. In some examples, the width D3 may be about 350 mm to about 675 mm. The top portion **154** of the hub **150** has a diameter or width D4 that may be smaller than the diameter or width D3 of the bottom portion **152**. The diameter or width D4 may be about 200 mm to about 600 mm. In some examples, the width D4 may be about 275 mm to about 475 mm. Larger and smaller dimensions are contemplated. In the depicted embodiment, there are three supports **156** connecting the bottom portion **152** to, and supporting, the top portion **154**. In some embodiments, there may be more or fewer than three supports **156**.

(24) The raised ring **138** of the heat deflector **102** is designed to receive the hub **150** and rest on the hub **150** instead of using the removable legs **104** described above with respect to FIGS. 1A and 1B. In this configuration, the hub **150** sits on the fire pit **106** with the bottom portion **152** of the hub **150** resting on the connecting ring **114**. The top lip **116** prevents the hub **150** from sliding off the fire pit **106**. The heat deflector **102** sits on top of the hub **150** with the top portion **154** of the hub **150** fitting inside the raised ring **138** of the heat deflector. In some embodiments, the weight of the heat deflector **102** maintains the connection between the heat deflector **102** and the hub **150**. The combined weight of the heat deflector **102** and the hub **150** maintain the connection between the hub **150** and the fire pit **106**. In some other embodiments, the heat deflector **102** may be fastened to the hub **150** and the hub **150** may be fastened to the fire pit **106**, such as with bolts or clips, for example. In some other embodiments, a combination of weight and fasteners may be used to install the heat deflector **102** on the fire pit **106**.

(25) FIG. 3 is a side view showing radiant energy deflected by the exemplary heat deflector **102** connected to an exemplary fire pit according to some embodiments of the present disclosure. In some embodiments, the heat deflector **102** is installed on the fire pit **106** using the removable legs **104**. In some other embodiments, the heat deflector **102** is installed on the fire pit **106** using the hub **150**. A person **302** is illustrated sitting next to the fire pit **106**. Radiant energy **304** (e.g., heat) is shown rising vertically from the fire pit **106** and being laterally deflected by the heat deflector **102** towards the person **302**. As discussed above, the angle α_1 of the outer portion **130** of the heat deflector **102** may be selected to provide a suitable angle of incidence of the strongest source of radiant energy **304**, and in some instance, maximizes the outer diameter or width of the fire pit **106** in relation to the specific location on the heat deflector **102**. Similarly, the angle α_3 of the inner portion **134** of the heat deflector **102** may be selected to provide suitable angle incidence of the strongest source of radiant energy **304**, the middle of the fire pit **106**, in relation to the specific location on the heat deflector **102**. By so doing, the heat deflector **102** efficiently, horizontally redirects the vertically moving radiant energy **304** toward the person **302**. Smoke **306** is shown passing through the plurality of holes **140** in the middle portion **134** of the heat deflector.

(26) FIGS. 4A-4E are views of an example heat deflector with the removable leg components, according to some embodiments of the present disclosure. Specifically, FIG. 4A illustrates a top-down view of the heat deflector **102**; FIG. 4B illustrates a perspective view of one removable leg **104**; FIG. 4C illustrates a perspective view of a cap **402** used to secure the removable leg **104** to the heat deflector **102**; FIG. 4D illustrates a perspective cross section view of the heat deflector **102** along line A-A without removable legs **104**; and FIG. 4E illustrates a perspective cross section view of the heat deflector **102** along line A-A with the removable legs **104** attached to the heat deflector **102** using caps **402**.

(27) As discussed above, the heat deflector **102** includes the main body with the outer portion **130**, the middle portion **132**, and the inner portion **134**. The outer portion, including the ribbed features **136**, has diameter or width D2. The middle portion **132**, including the plurality of holes **140** and the raised ring **138**, has a diameter or width D5 that may be about 200 mm to about 600 mm. In some examples, the width D5 may be about 250 mm to about 500 mm. The inner portion has a diameter or width D6 that may be about 100 mm to about 350 mm. In some examples, the width D6 may be about 150 mm to about 225 mm. Although larger and smaller widths are contemplated. The outer edge **160** of the outer portion **130**, and thus the heat deflector **102**, is folded over as illustrated in FIGS. 4D and 4E to form an edge lip. The folded structure of the outer edge **160** improves the structural and aesthetic of the heat deflector **102**. The outer edge **160** improves the structural strength of the heat deflector **102** to resist warping and bending during repeated heating and cooling cycles. The outer edge **160** further provides a clean look to the heat deflector **102** as well as protects against sharp edges.

(28) FIGS. 4D and 4E illustrate differences in angle α_1 of the outer portion **130**, angle α_2 of the ribbed features **136**, and angle α_3 of the inner portion **134**. As described above, the angles of the outer portion **130** and the inner portion **134** maximize the angle of incidence of the radiant energy rising from the fire pit **106**.

(29) FIG. 4B illustrates a perspective view of a removable leg **104**. The removable leg **104** includes the upper portion **105**, the lower portion **107**, an upper support **404**, and support features **406**. The upper portion **105** may include one or more posts, or legs, that pass through the connection point **122** of the heat deflector **102**. The upper portion **105** may lock, or snap, into the heat deflector. Alternatively, the cap **402**, such as illustrated in FIG. 4C, may snap, or lock, onto the upper portion **105** of the removable leg **104** to secure the removable leg **104** to the heat deflector **102**. The cap **402** may be curved to match the curve of the raised ring **138** to sit flush with the raised ring **138** of the heat deflector **102**. The cap **402** snaps, or locks, onto the upper portion **105** of the removable leg **104**. The bottom portion **107** of the removable leg **104** provides a footing for the heat deflector **102** when the removable leg **104** is installed in the heat deflector **102**. The bottom portion **107** contacts the ground, the connecting ring **114**, or other flat surface supporting the heat deflector **102**. The upper support **404** interfaces with heat deflector **102** when the removable leg **104** is installed on the heat deflector **102**.

(30) FIGS. 4D and 4E further illustrate a cross section of the raised ring **138** having an outer vertical portion **410**, a top horizontal portion **412**, and a sloped inner portion **414**. The outer vertical portion **410** and top horizontal portion **412** allow the heat deflector **102** to rest on the top portion **154** of the hub **150**, as described above. The sloped inner portion **414** includes the connection points **122** to receive the upper portion **105** of removable leg **104**. A catch feature **416** of the ribbed feature **136** receives the upper support **404** of the removable leg **104** during installation and while installed. The catch feature **416** may be a bent portion of the ring that allows the leg to connect thereto.

(31) FIGS. 5A-5C and 6A-6C illustrate inserting the removable leg **104** into the heat deflector **102** and using the cap **402** to secure the removable leg **104** to the heat deflector **102**. The process begins in FIG. 5A with a removable leg **104** and a heat deflector **102**. The bottom portion **107** of the removable leg **104** is angled away from the center of the heat deflector **102** while the upper portion **105** of the removable leg **104** is angled toward the center of the heat deflector **102**. The removable leg **104** is moved toward the connection point **122** and the catch feature **416** located on the underside of the heat deflector **102**, specifically near the ribbed feature **136**. At FIG. 5B, the upper support **404** of the removable leg **104** engages the catch feature **416**, allowing the removable leg **104** to pivot about the catch feature **416**. This places the upper portion **105** of the removable leg **104** adjacent the connection point **122**. The removable leg **104** is then rotated clockwise, as indicated in FIG. 5B, so that the upper portion **105** of the removable leg **104** extends through the connection point **122** of the heat deflector **102**, as shown in FIG. 6A. The upper support **404** maintains contact with the catch feature **416** throughout the rotation of the removable leg **104** thereby securing the removable leg **104** from falling down or over rotating during installation. As illustrated in FIG. 5C, the removable leg **104** is now in position within the heat deflector **102**. A similar process is repeated for each of the removable legs **104** to be attached to the heat deflector **102**. At this point, the heat deflector **102** may be able to stand using the removable legs **104**.

(32) The installation process continues as shown in FIG. 6A where a cap 402 is placed over the of the upper portion 105 of the removable leg 104. After placing the cap 402 over the upper portion 105, the cap 402 slides toward the center of the heat deflector 102 as illustrated in FIG. 6B. Sliding the cap 402 over the upper portion 105 engages the cap 402 to the upper portion 105 of the removable leg 104 preventing the removable leg 104 from rotating out of the connection point 122 and falling off the heat deflector 102. FIG. 6C shows the removable leg 104 completely installed in the heat deflector 102. The same process is repeated for each of the remaining removable legs 104.

(33) FIG. 7 shows a bottom view of an exemplary heat deflector. The heat deflector 102 includes the outer portion 130, the middle portion 132, and the inner portion 134. The outer edge of the outer portion 130 includes the fold of the outer edge 160. The ribbed features 136 are seen from beneath the heat deflector 102. Additionally, the catch feature 416 is seen from beneath the heat deflector 102. The middle portion 132 includes the raised ring 138 and the plurality of holes 140 as described above. The inner portion 134 includes the hole 142 at the vertex of the conical shape.

(34) A number of variations are possible on the examples and embodiments described herein. For example, the fire pit, heat deflector, removable legs, hub, or other components could be made of heavier-gauge material in order to support more weight, or of lighter gauge material in order to become lighter and more portable. The heat deflector could be made in different sizes and/or with different degrees of curvature. Air gaps may be larger or smaller than shown herein, to optimize air flow through the fire pit, to minimize weight or volume of the fire pit, or for other reasons. The relative lengths, widths, and radii of different components could be different than presented herein. The fire pit, heat deflector, removable legs, hub, or other components could be made by different processes, including casting, forging, sintering, milling, or 3D printing. They could be made of different metals, or of nonmetallic materials such as ceramics. The fire pit rim could be noncircular, including such possible shapes as ovals, rectangles, triangles, and rhombuses. The technology described herein may be used to burn firewood, wood chips or pellets, scrap lumber, paper, cardboard, coal, and other combustible materials. It may be employed for example in lamps, stoves, fire pits, fireplaces, furnaces, forges, and boilers, and other combustion heaters. In some implementations, the fire pit, heat deflector, removable legs, hub, or other components may comprise several pieces that collectively form a structure like that described herein.

(35) The logical operations making up the embodiments of the technology described herein are referred to variously as operations, steps, objects, elements, components, or modules. Furthermore, it should be understood that these may occur or be performed in any order, unless explicitly claimed otherwise or a specific order is inherently necessitated by the claim language.

(36) Attached hereto is an Appendix that includes Figures A through K. Specifically, in several embodiments, one or more of the embodiments of the present application are provided in whole or in part as described and illustrated in the Appendix, which forms part of the present application. Moreover, Figures A through K provide additional support for any U.S. or non-U.S. design applications that are to be filed in the future claiming priority to this present U.S. utility patent application. More particularly, in the Appendix:

(37) FIG. A is a top perspective view of a new, original design for a DEVICE FOR FIRE PIT according to a first embodiment;

(38) FIG. B is a top view of the first embodiment;

(39) FIG. C is a bottom view of the first embodiment;

(40) FIG. D is a front view of the first embodiment;

(41) FIG. E is a rear view of the first embodiment;

(42) FIG. F is a left side view of the first embodiment;

(43) FIG. G is a right side view of the first embodiment; and

(44) FIG. H is a bottom perspective view of the first embodiment.

(45) FIG. 1 is a top perspective view of the new, original design for a DEVICE FOR

(46) FIRE PIT according to a second embodiment, the second embodiment being identical to the first embodiment except that the second embodiment has an indeterminate diameter as indicated in FIG. 1.

(47) FIG. J is a view similar to FIG. D but with additional broken lines.

(48) FIG. K is a view similar to FIG. D but with additional broken lines.

(49) In several embodiments, one or more of the embodiments described and illustrated in the Appendix are combined in whole or in part with one or more of the embodiments described above, illustrated in one or more of FIGS. 1 through 16, one or more other embodiments described and illustrated in the Appendix, or any combination thereof.

(50) All directional references e.g., upper, lower, inner, outer, upward, downward, left, right, lateral, front, back, top, bottom, above, below, vertical, horizontal, clockwise, counterclockwise, proximal, and distal are only used for identification purposes to aid the reader's understanding of the claimed subject matter, and do not create limitations, particularly as to the position, orientation, or use of the fire grate, ash pan, or fire pit. Connection references, e.g., attached, coupled, connected, and joined are to be construed broadly and may include intermediate members between a collection of elements and relative movement between elements unless otherwise indicated. As such, connection references do not necessarily imply that two elements are directly connected and in fixed relation to each other. The term “or” shall be interpreted to mean “and/or” rather than “exclusive or.” Unless otherwise noted in the claims, stated values shall be interpreted as illustrative only and shall not be taken to be limiting.

(51) The above specification, examples and data provide a complete description of the structure and use of exemplary embodiments of the fire grate, ash pan, and fire pit as defined in the claims. Although various embodiments of the claimed subject matter have been described above with a certain degree of particularity, or with reference to one or more individual embodiments, those skilled in the art could make numerous alterations to the disclosed embodiments without departing from the spirit or scope of the claimed subject matter. Still other embodiments are contemplated. It is intended that all matter contained in the above description and shown in the accompanying drawings shall be interpreted as illustrative only of particular embodiments and not limiting. Changes in detail or structure may be made without departing from the basic elements of the subject matter as defined in the following claims.

Claims

1. A heat deflector for use with a fire pit having a burn chamber defined by a fire pit sidewall, the burn chamber having a chamber width, the heat deflector comprising: a deflection panel including: a first region disposed at a first angle with respect to the horizontal, wherein the first region includes a plurality of ventilation holes, the first region being configured to be disposed directly above the burn chamber when in use; a second region surrounding the first region and extending radially outwardly, the second region being disposed at second angle with respect to the horizontal, the second region being configured to be disposed directly over the fire pit sidewall when in use and aligned such that an outer edge of the second region is radially outward from the sidewall and an inner edge of the second region is radially inward from the sidewall, wherein the second angle is different than the first angle; and a central region surrounded by the first region and disposed at a third angle with respect to the horizontal, the central region being configured to be centrally disposed directly over the burn chamber, wherein the third angle is different than the first angle; and a support configured to maintain the deflection panel in a position over the fire pit.
2. The heat deflector of claim 1, wherein the support comprises a plurality of support legs configured to extend from the heat deflection panel to the fire pit, wherein the legs are attached to the heat deflection panel at a location radially outward of the first region and the central region.
3. The heat deflector of claim 2, wherein the support legs are removable.
4. The heat deflector of claim 2, wherein the legs are attached to the heat deflection panel with fasteners.
5. The heat deflector of claim 1, wherein the heat deflection panel comprises a plurality of pieces joined by fasteners.
6. The heat deflector of claim 5, wherein at least some of the pieces are formed by stamping a sheet of metal.

7. The heat deflector of claim 5, wherein at least some of the pieces are formed by folding.
 8. The heat deflector of claim 1, further comprising a plurality of stiffening features disposed on the second region.
 9. The heat deflector of claim 8, wherein the stiffening features are formed by stamping.
 10. The heat deflector of claim 9, wherein the stiffening features are distributed non-equidistantly around the circumference of the second region.
 11. A heat deflector for use with a fire pit having a burn chamber defined by a fire pit sidewall, the burn chamber having a chamber width, the heat deflector comprising: a deflection panel including: an inner region having a hollow truncated cone shape in cross-section, the cone shape having a base with a base width measured at the base, the cone shape having a flat vertex surface, and a first heat deflecting surface extending from the flat vertex surface to the base, wherein the flat vertex surface is below the base, the inner region being configured for suspension directly over the burn chamber when in use, the base width being smaller than the chamber width; and an outer region disposed radially outward of the inner region, the outer region having an outer edge portion and an inner edge portion, the outer edge having a second heat deflecting surface disposed between the outer edge portion and the inner edge portion, the second heat deflecting surface of the outer region being configured to be disposed directly over the fire pit sidewall defining the burn chamber of the fire pit when in use and aligned such that the outer edge portion is radially outward from the sidewall and the inner edge portion is radially inward from the sidewall; and a support configured to maintain the deflection panel in a position over the fire pit when in use.
 12. The heat deflector of claim 11, wherein the inner edge portion of the outer region is disposed at an elevation that is about the same as an elevation of the base of the truncated cone.
 13. The heat deflector of claim 11, further comprising a through hole disposed in the flat vertex surface.
 14. The heat deflector of claim 11, wherein the deflection panel comprises a ring of ventilation holes disposed so as to be directly above the burn chamber when in use, the ventilation holes being configured to permit passage of smoke.
 15. The heat deflector of claim 11, wherein the support comprises a plurality of support legs configured to extend from the heat deflection panel to the fire pit, wherein the legs are attached to the heat deflection panel at a location radially outward of the inner region.
 16. A heat deflector for use with a fire pit having a burn chamber defined by a fire pit sidewall, the burn chamber having a chamber width, the heat deflector comprising: a deflection panel including: a burn chamber portion defined by a vertical reference boundary of the burn chamber of the fire pit when the heat deflector is in use with the fire pit, the burn chamber portion comprising a first heat deflecting surface forming a cone-shape in cross-section, the first heat deflecting surface being configured to deflect heat in a lateral direction, the burn chamber portion having a plurality of ventilation holes sized to permit passage of smoke from the burn chamber therethrough; and a peripheral portion outside the vertical reference boundary of the burn chamber of the fire pit when the heat deflector is in use with the fire pit, the peripheral portion comprising a second heat deflecting surface configured to deflect heat in a lateral direction, the second heat deflecting surface being angled differently than the first heat deflecting surface and aligned such that an outer edge of the peripheral portion is radially outward from the sidewall and an inner edge of the peripheral portion is radially inward from the sidewall; and a support configured to maintain the deflection panel in a position over the fire pit when in use.
 17. The heat deflector of claim 16, wherein the ventilation holes are formed in a ring-shaped portion of the burn chamber portion.
 18. The heat deflector of claim 16, wherein the burn chamber portion forming a cone-shape in cross-section includes a base edge and includes a flat vertex surface, wherein the flat vertex surface is below the base edge.
 19. The heat deflector of claim 18, further comprising a through hole disposed in the flat vertex surface.
 20. The heat deflector of claim 16, wherein the support comprises a plurality of support legs configured to extend from the heat deflection panel to the fire pit when in use with the fire pit.
 21. The heat deflector of claim 16, wherein the support comprises a hub configured to extend from the heat deflection panel to the fire pit when in use with the fire pit.
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